

WARROAD, MN, USA

WMO: 726548

Lat: 48.941N

Lon: 95.348W

Elev: 1074

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04971

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-18.2	-32.7	0.9	-23.0	-26.4	1.3	-18.5	22.5	13.4	20.1	14.2	4.8	260	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	84.3	72.5	81.8	70.3	79.4	67.9	75.1	82.1	72.7	79.8	70.4	77.6	10.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	125.6	79.9	70.3	116.1	77.9	67.7	106.0	75.7	39.3	82.4	37.0	79.7	34.9	78.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.9	17.0	DB	-30.6	89.9	6.2	2.8	-35.1	91.9	-38.8	93.6	-42.2	95.2	-46.8	97.2	(4)
(5)			WB	-30.7	78.6	5.9	3.5	-35.0	81.1	-38.4	83.1	-41.7	85.1	-46.0	87.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	6.0	9.5	24.3	38.8	51.7	62.5	67.4	64.9	56.6	43.1	28.0	12.9		
	DBStd	23.83	15.05	13.75	13.18	9.90	8.74	6.53	5.73	6.19	8.33	9.07	11.13	13.46		
	HDD50	5831	1364	1135	800	352	86	2	0	1	29	253	660	1149		
	HDD65	9769	1829	1555	1263	785	421	121	41	78	272	681	1109	1614		
	CDD50	1805	0	0	2	18	137	378	540	463	228	38	1	0		
	CDD65	266	0	0	0	0	7	47	116	74	21	1	0	0		
	CDH74	2227	0	0	0	4	82	399	946	599	189	8	0	0		
	CDH80	490	0	0	0	0	13	92	214	131	39	1	0	0		
	WSAvg	7.9	7.5	7.6	7.9	8.7	9.0	8.0	7.1	7.0	7.9	8.5	8.4	7.5		
	PrecAvg	25.4	0.8	0.7	0.9	1.4	3.2	4.7	3.6	3.3	2.9	2.1	1.1	0.9		
(15) Precipitation	PrecMax	33.5	1.8	2.1	2.9	3.1	7.6	12.7	5.6	6.6	7.8	5.2	3.9	1.6		
	PrecMin	17.4	0.2	0.0	0.2	0.3	1.4	1.9	1.7	0.9	0.6	0.3	0.2	0.4		
	PrecStd	4.3	0.4	0.5	0.7	0.6	1.7	2.6	1.2	1.4	1.9	1.3	0.9	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	43.0	61.4	72.4	81.4	88.0	89.6	88.2	84.2	73.2	59.0	40.9		
		MCWB	36.2	38.8	51.5	56.4	63.6	73.0	75.5	75.9	71.1	61.1	50.5	36.4		
	2%	DB	35.5	37.1	50.2	64.3	75.1	82.0	84.3	83.6	79.5	66.4	52.0	36.6		
		MCWB	33.2	34.6	44.5	51.5	60.9	69.5	72.9	73.2	67.5	57.9	46.4	34.0		
	5%	DB	30.3	34.3	45.0	60.6	71.8	79.1	81.9	80.9	74.7	62.6	46.2	34.2		
		MCWB	29.3	32.1	40.0	49.1	58.8	66.7	71.2	70.1	64.8	54.4	42.2	31.9		
	10%	DB	27.2	30.0	39.5	54.4	66.2	74.8	79.4	77.1	71.6	56.7	43.0	30.3		
		MCWB	25.6	28.9	35.9	45.6	56.5	64.9	68.9	67.1	63.1	50.6	38.9	29.3		
	0.4%	WB	36.9	39.0	53.2	57.6	68.2	76.0	78.2	78.2	74.1	63.8	52.2	37.4		
		MCDB	40.1	41.5	60.0	67.2	76.8	82.5	84.9	85.7	82.8	69.7	57.0	40.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.4	35.5	44.6	53.0	63.8	72.2	75.5	74.4	69.4	59.0	46.4	34.3		
		MCDB	34.3	37.5	49.2	63.1	71.5	80.6	82.3	81.3	76.7	66.2	49.9	36.7		
	5%	WB	29.1	32.1	40.0	49.8	60.3	68.6	73.2	71.4	66.3	54.6	42.6	31.9		
		MCDB	30.0	34.1	44.5	58.7	67.3	75.9	79.6	78.7	73.4	60.8	46.7	33.9		
	10%	WB	25.3	28.6	36.9	46.3	57.9	66.2	71.0	68.9	63.7	50.9	38.9	29.1		
		MCDB	27.2	29.5	40.4	53.6	65.9	73.4	77.9	75.6	69.5	56.3	42.5	30.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	18.6	18.0	18.4	18.9	18.0	19.1	19.3	19.0	15.6	13.4	14.5		
		MCDBR	16.4	16.6	19.5	26.7	25.7	22.9	21.5	22.7	24.1	23.0	18.0	15.8		
		MCWBR	15.0	14.1	14.6	16.7	14.6	13.4	13.2	13.9	14.8	15.0	13.4	13.9		
	5% WB	MCDBR	16.5	15.2	18.8	24.6	21.0	20.0	19.3	20.4	21.4	20.9	17.3	15.3		
		MCWBR	15.2	13.1	14.4	16.3	13.7	13.2	13.1	13.9	14.3	14.7	13.2	13.7		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.293	0.294	0.344	0.360	0.376	0.380	0.374	0.346	0.304	0.275	0.254		
	taud		2.292	2.200	2.398	2.382	2.401	2.383	2.390	2.417	2.483	2.568	2.486	2.357		
	Ebn at Noon		256	270	294	287	285	280	277	274	272	269	253	244		
	Edh at Noon		22	31	31	35	36	37	36	33	28	22	19	18		
(44) All-Sky Solar Radiation	RadAvg		435	814	1193	1516	1705	1791	1914	1594	1178	690	386	311		
	RadStd		47	65	118	138	110	142	94	117	96	100	46	34		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page